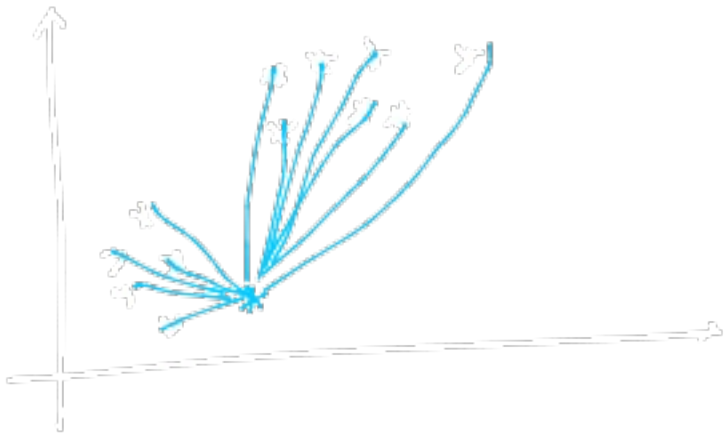
Solu

(i) distance → dimension → 2D → nD

G1 --- G000

(i) income
(ii) credit_score

historical data → plot it



LA

(ii) line

	Income	C-S	Status
C1	17	750	Yes
C2	16	450	No

dimension $\boxed{2D}$

dimension ↑

dimension

↓

vis. → math

↑

linear Alge → $\boxed{3D}$

	in	cibil-score	status
c_1	—	—	
c_2	—	—	
c_3	—	—	
c_4	—	—	
c_5	—	—	

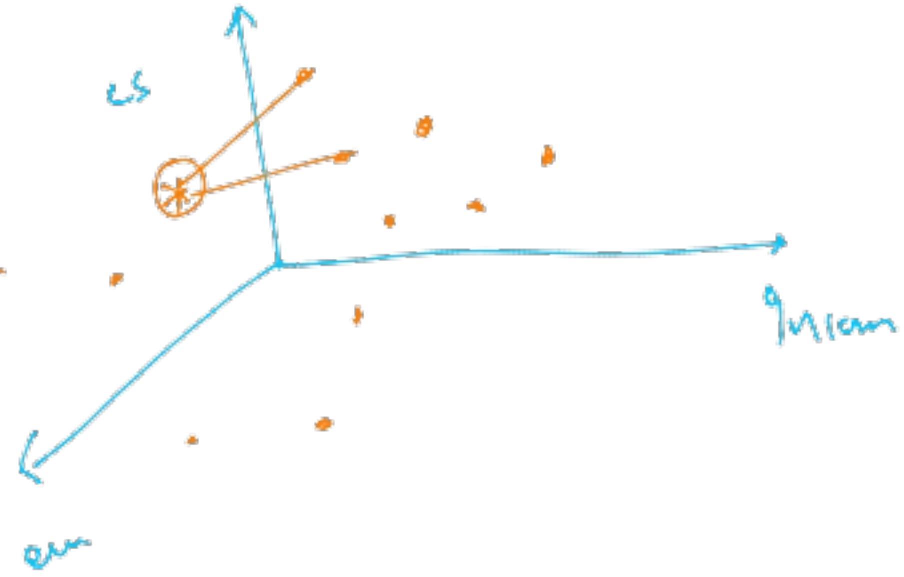
1M ← 2D

2^n

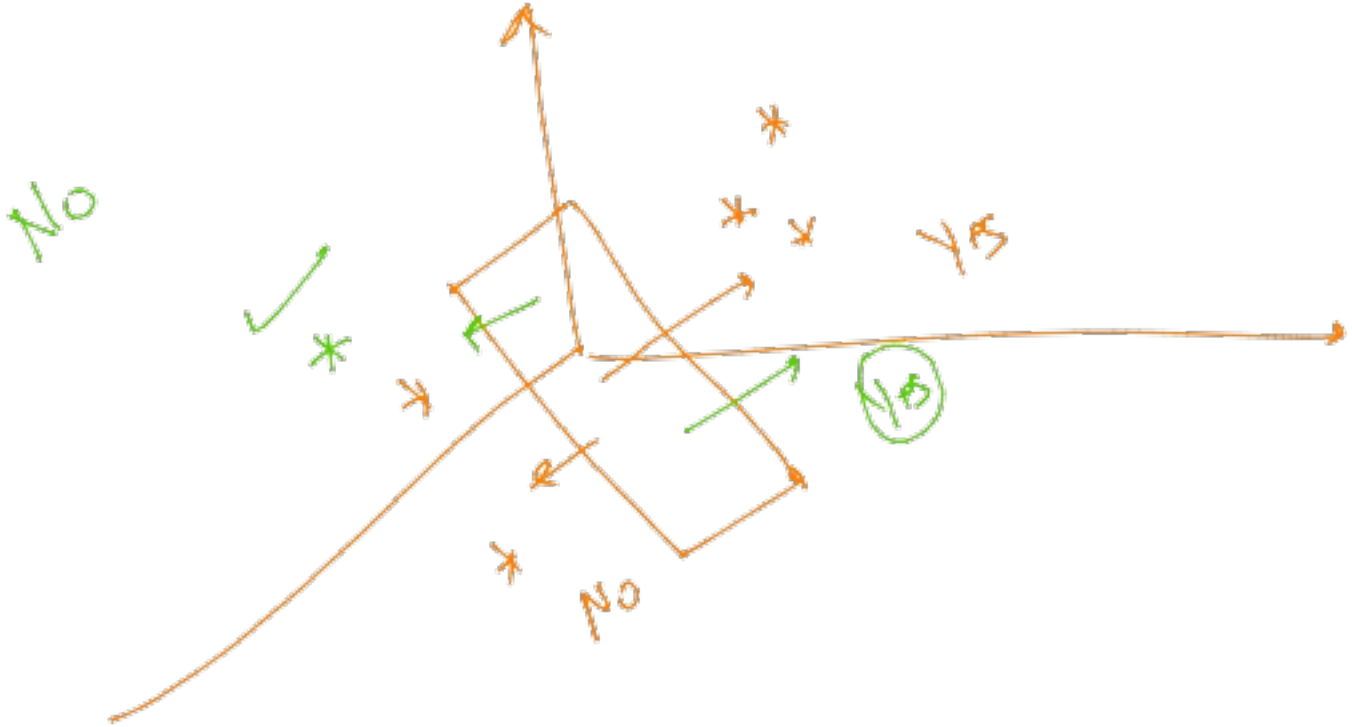
i cs employee-year

C1 (– – 3) } Historical

C2 C



if you move to higher dimensions ↓



2D → line

3D → plane

4D

nD → Hyperplane

maths

in CS Age Exp — — — —

C1

⇓

⇓

Viz But you can do

Yes ← C1 (17, 750, 7, 16)
Yes ← C2 (16, 780, 7, 15)
Yes ← C3 (16, 790, 6, 14)
No ← C4 (9, 390, 5, 4)
No ← C5 (4, 380, 5, 6)

But

Yes → 3
No → 0
→

C_New ⇒ [16, 580, 7, 14] → ? < Yes / No

Yes

New close

3

C_New
C1 → 0.75
C2 → 0.65
C3 → 0.55
C4 → 0.95
C5 → 1.05

Yes

C_New → distance(C_New, C1) = $\sqrt{(-1)^2 + (750-580)^2 + (7-7)^2 + (2)^2}$

= 0.75

↳ distance(C_New, C2)

(i) distance
(ii) line

Yes
No

} classification problem

New

Retail domain
⇓
$\boxed{\text{data}}$

But

ABC

|| week

day	discount	B1G1	Holiday-flag	sale-volume
1	0.2	F	F	350
2	0.2	T	F	380
3	0.25	F	F	370
4	0	T	T	270
5	—	—	—	—
6	—	—	—	—
7	—	—	—	—

7-days

Help me to build some strategy to increase sale-volume ↑

8th

Plan

↓

promotion

0.15

(~~0.25~~, T, T) → logic → Sales-volume ?

Input

↑

output

(ii) Regression

Prediction → Sales-volume

↳ ∈ Real

210, 271 or ?

(i) classification
(ii) Regression

→ (i) maths

↓

(A) Distance

(B) Line/ plane/ Hyperplane

Forecasting

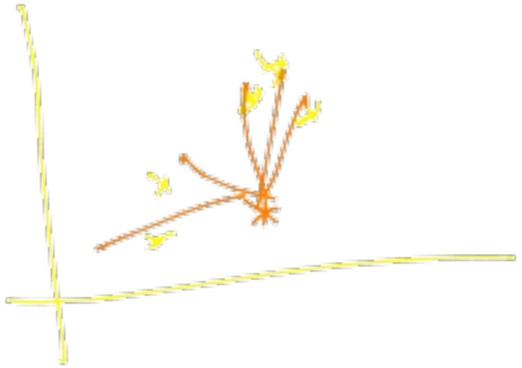
classification

⇓

Regre

maths = = ML

		X		Y
	Day	discount	BlGl	sales-volum
P_1	1	(0.25	T)	370
P_2	2	(0.10	T)	200
P_3	3	(0.15	F)	150
P_4	4	(0.16	F)	156
P_5	5	(0.15	F)	158
	8	[0.05	F]	?



distance

pick → 3

$$\text{Avg} = \frac{150+156+158}{3} \Rightarrow 155$$

Data

Historical ← collect

↓

Problem statement ?

↓

Type of problem →

(i) classification

(ii) Regression

DE

↓

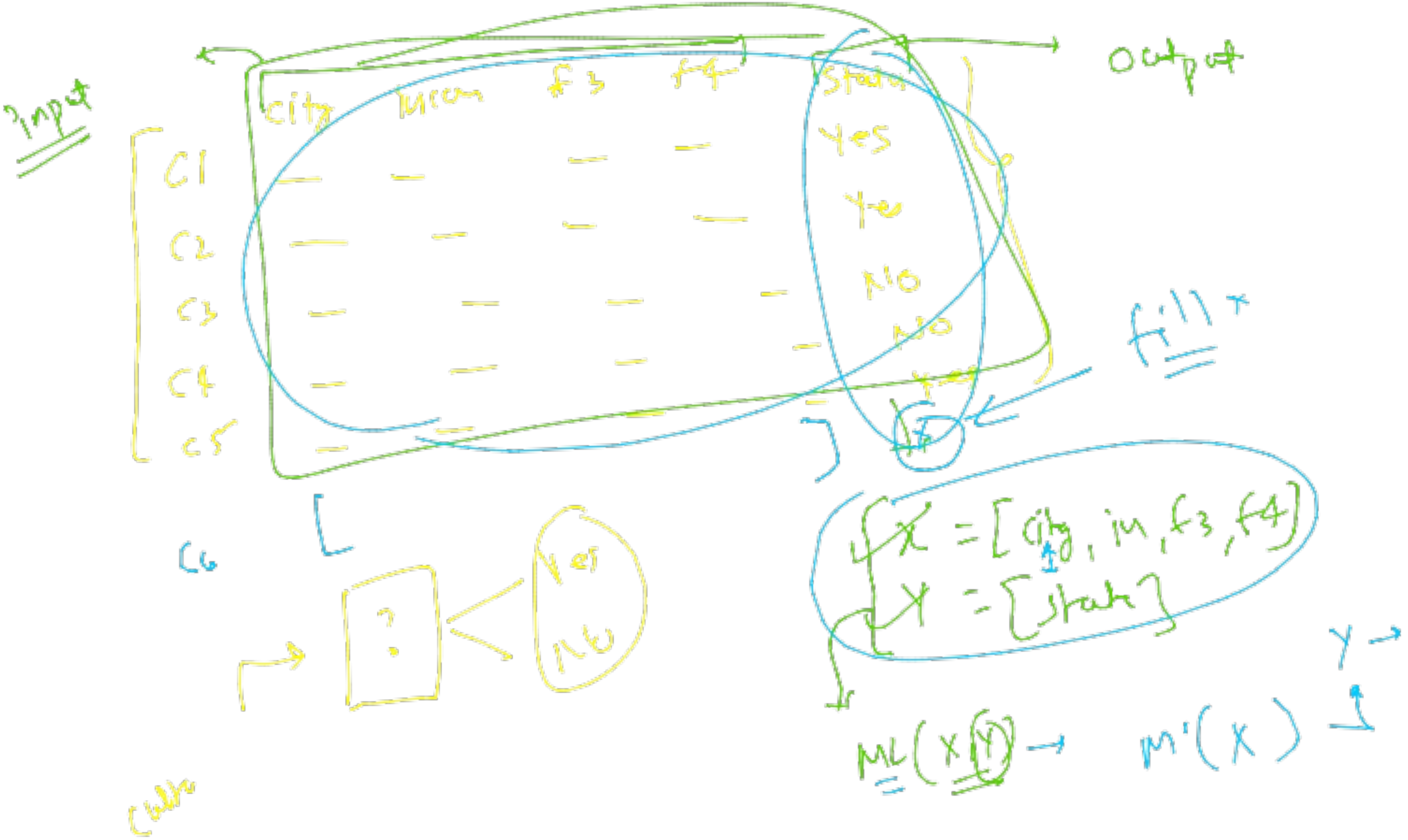
Raw

↓

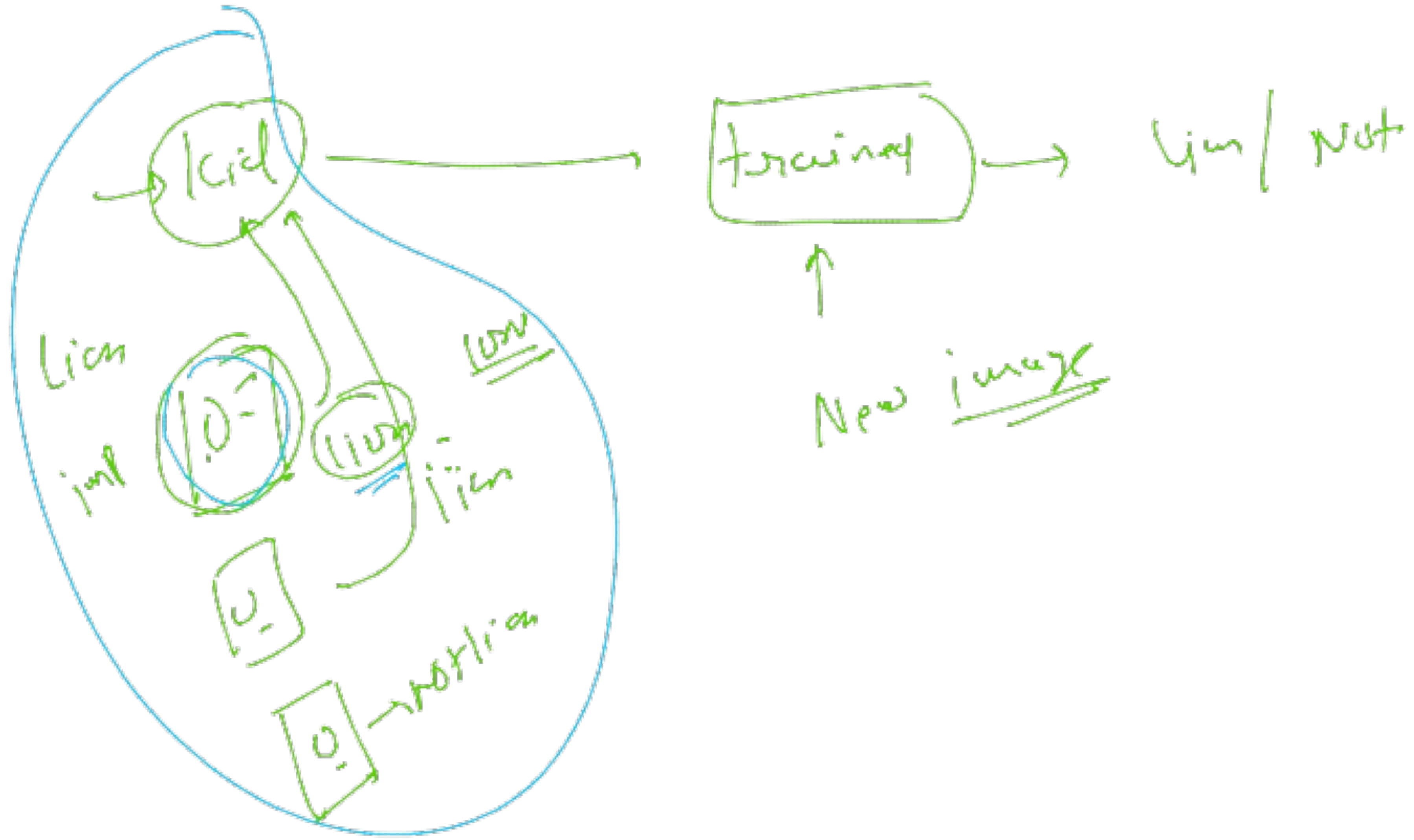
pre-processing

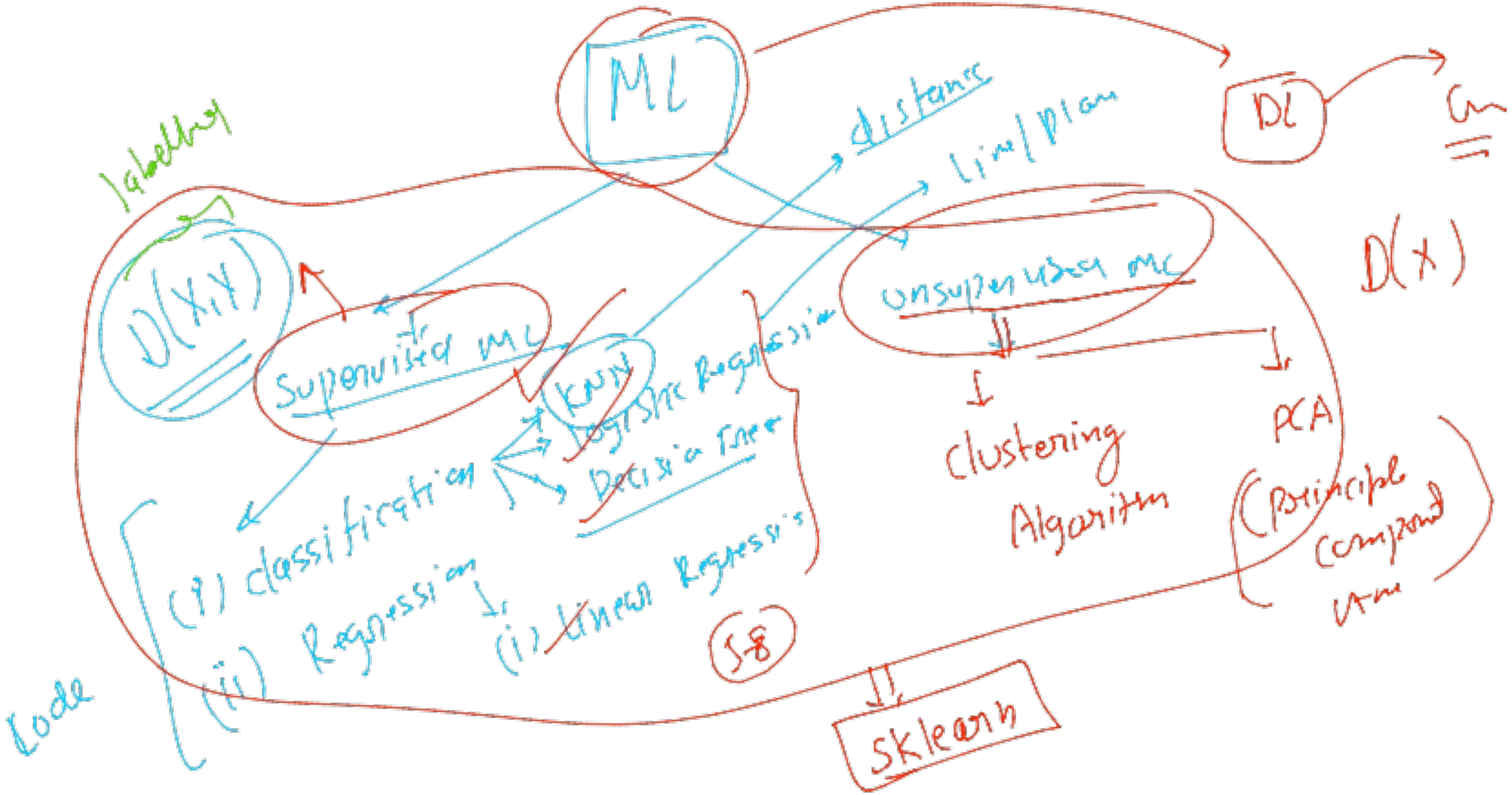
↓

Final table



Input
city
incm
f3
f4
Status
Output
C1
C2
C3
C4
C5
Yes
Yes
No
No
Yes
fill
C6
?
Yes
No
X = [city, in, f3, f4]
Y = [Status]
ML(X, Y) →
M'(X)
Y →


training
New image


ML
DL
distance
line/plane
labelling
D(X,Y)
Supervised ML
KNN
Logistic Regression
Decision Tree
Unsupervised ML
D(X)
Clustering Algorithm
PCA
(Principle component
(i) classification
(ii) Regression
(i) Linear Regression
Code
Sklearn

prep

Book

case

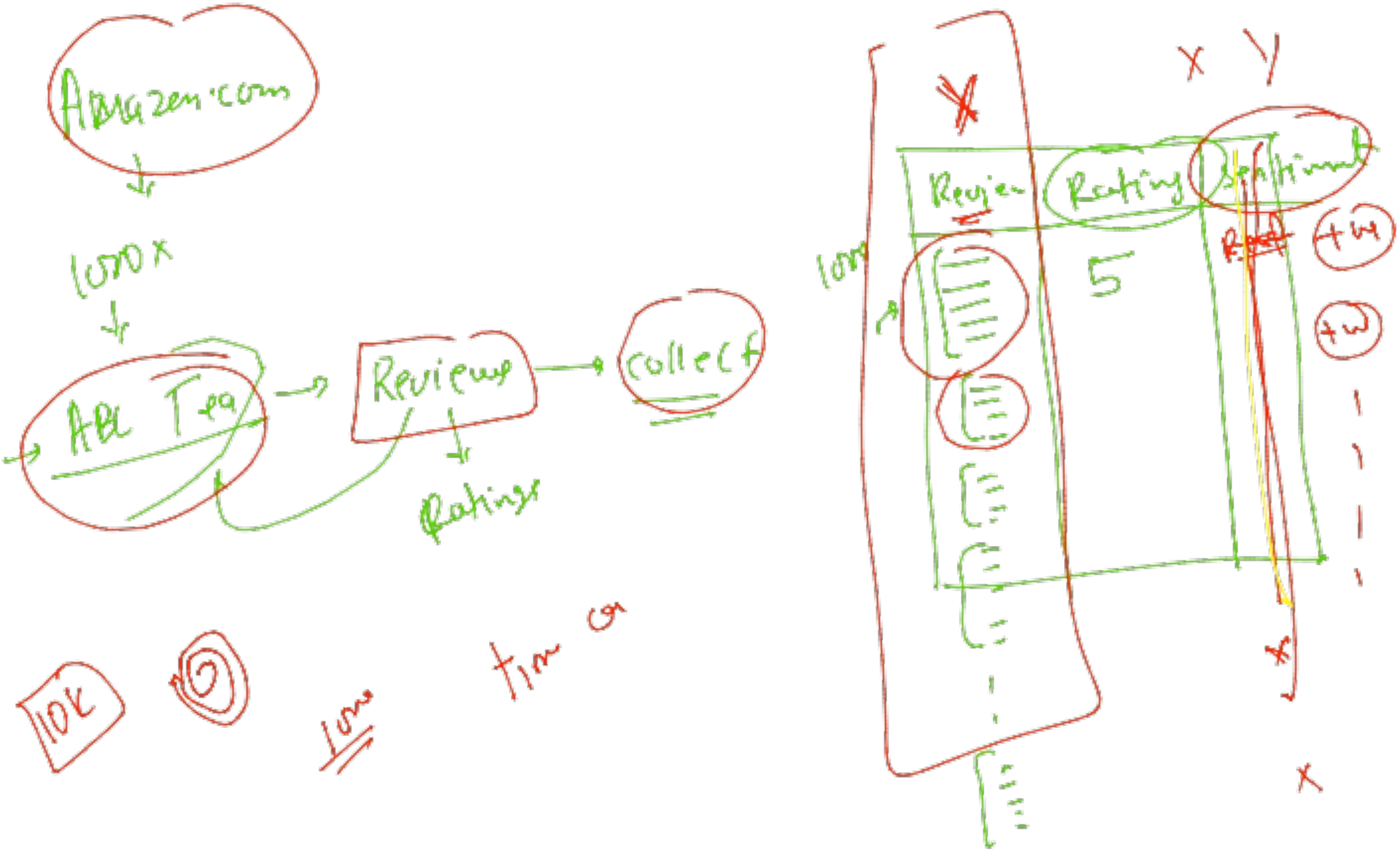
Q - A

Q → Exam → A

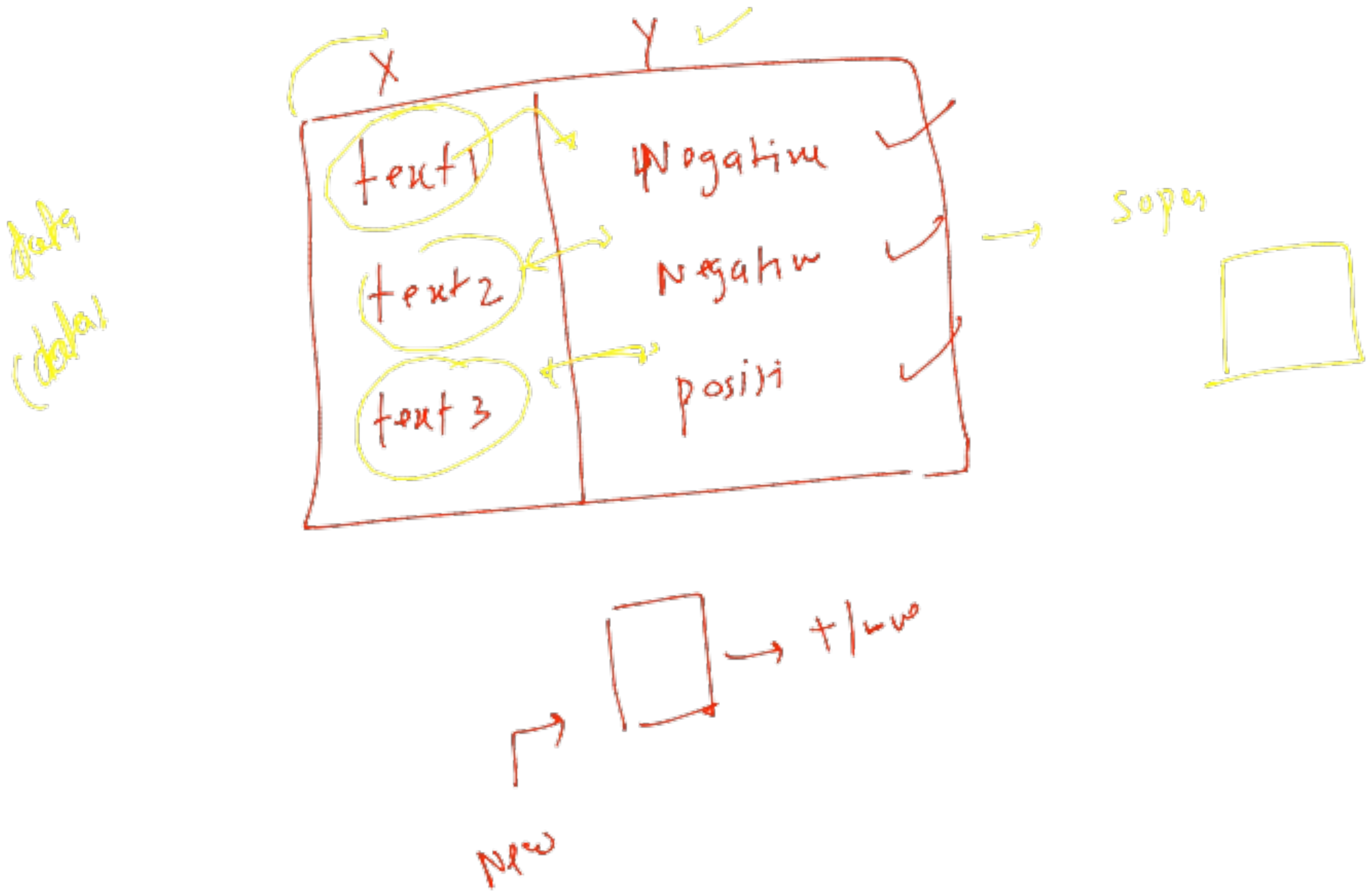
⇒ Book

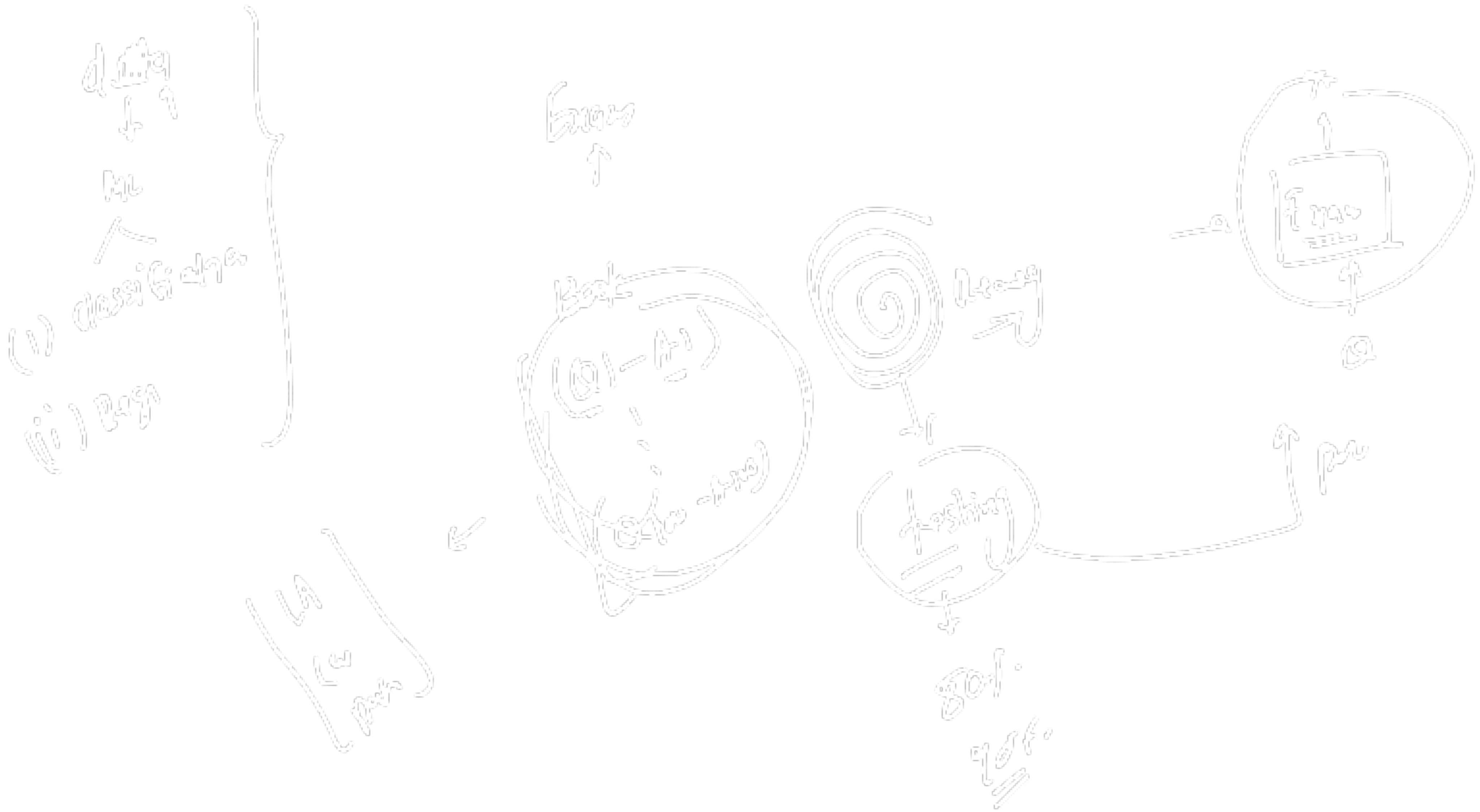
(x)

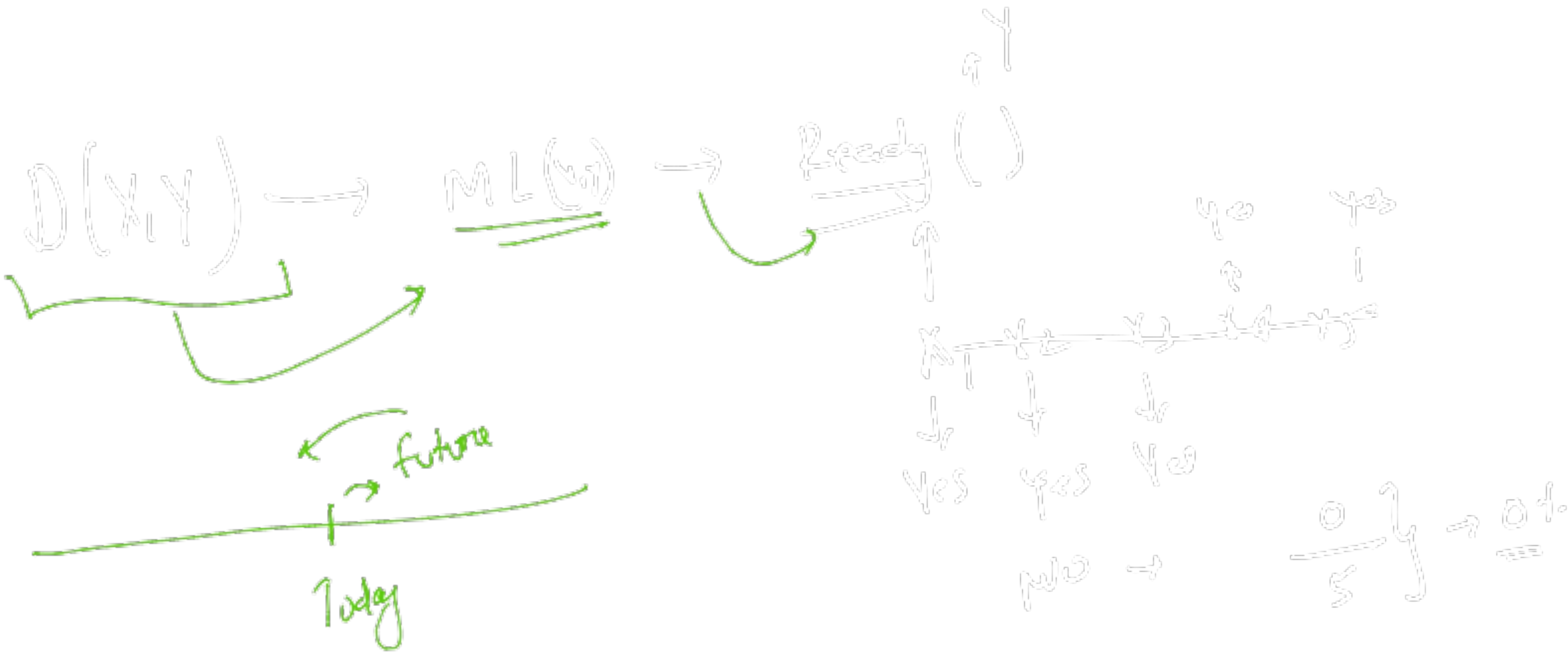
eg.

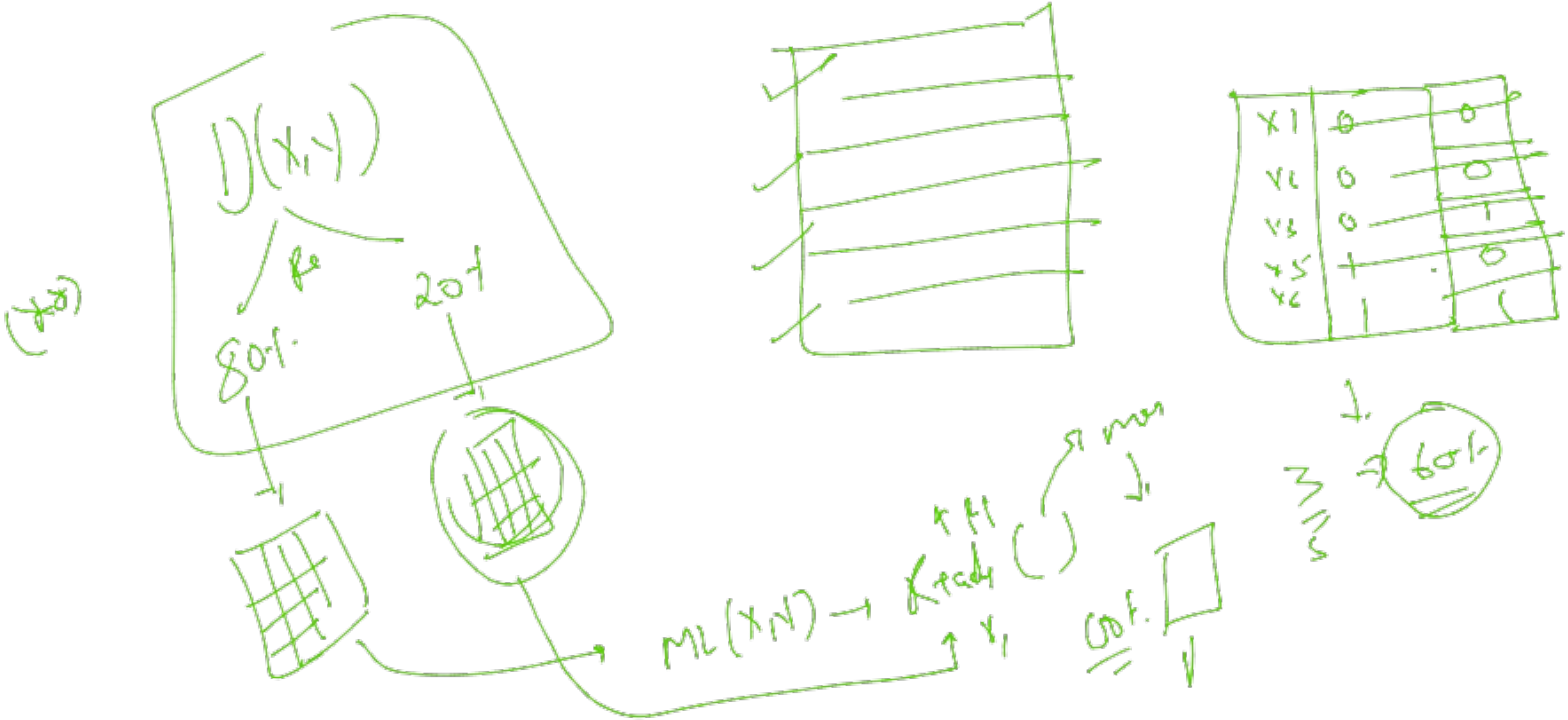


Amazon.com
1000x
ABC Tea
Reviews
Ratings
collect
X
Y
Review
Rating
Sentiment
5
1000
+ve
+ve
10k
1000
time


X
Y
text1
Negative
Negative
text2
text3
positi
data
(data)
+/-ve
New


data
ML
(i) classification
(ii) Regr
Exam
Back
Learning
testing
80%
Error

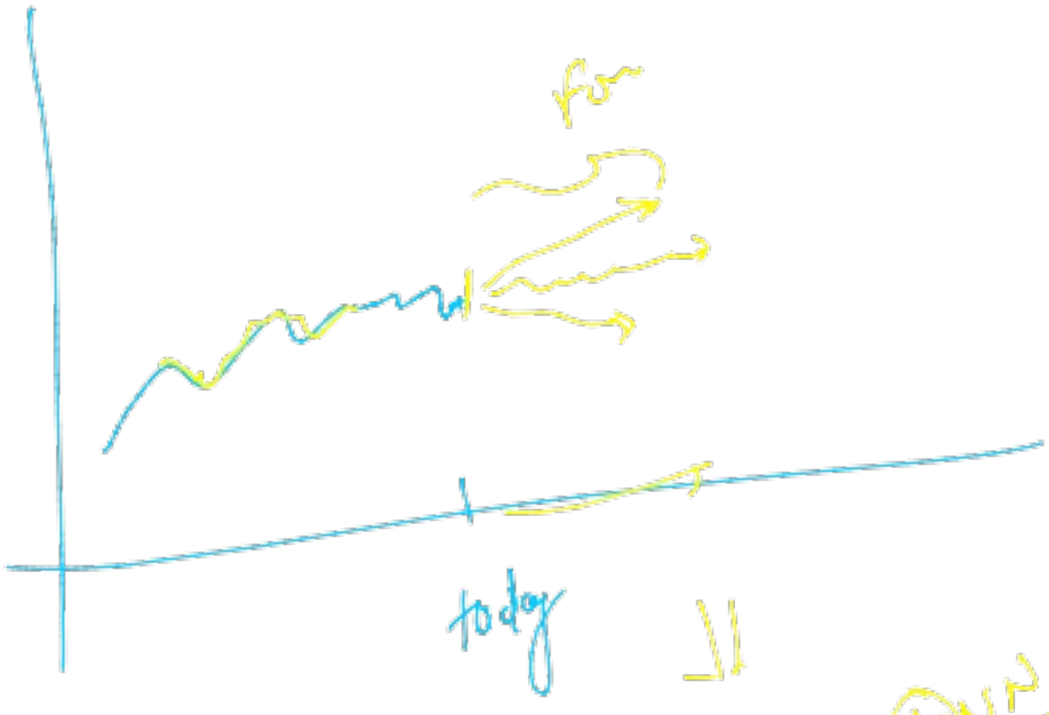

D(X,Y) → ML(X,Y) → Predict()
future
Today
yes
no


D(X,Y)
80%
20%
ML(X,Y)
X1
60%

ARIMA
STATIONARY

Forecasting

Reg



DL → RNN

Bulk

+

1000

→ 80 - 90H.

→ 80-90 DH.

250

Test

800

Data

↓

'EDA'

↓

split → testing

↓

training

↓

alg(X,y)

↓

output →